United States Patent	12415306
Kind Code	B2
Date of Patent	September 16, 2025
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Blow-molding mold, bottle blowing machine and plastic bottle

Abstract

Provided are a blow-molding mold, a bottle blowing machine and a plastic bottle, the blow-molding mold has a mold cavity, where a bottle body, a bottle shoulder, a handle and a bottle neck are formed within the mold cavity after a bottle preform is blown, the bottle shoulder is located between the bottle body and the bottle neck, one end of the handle is connected to the bottle shoulder, the other end of the handle is connected to the bottle shoulder or the bottle neck, and the mold cavity includes a first concave region to have a raised step structure formed on the bottle neck.

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Appl. No.:	**18/020551**
Filed (or PCT Filed):	**September 29, 2020**
PCT No.:	**PCT/CN2020/118769**
PCT Pub. No.:	**WO2022/056960**
PCT Pub. Date:	March 24, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230302711 A1	Sep. 28, 2023

Foreign Application Priority Data

CN	202010979443.5	Sep. 17, 2020
CN	202022045751.9	Sep. 17, 2020

Publication Classification

Int. Cl.: **B29C33/42** (20060101); **B29C49/48** (20060101); B29K67/00 (20060101); B29L31/00 (20060101)

U.S. Cl.:

CPC **B29C49/482** (20130101); **B29C33/42** (20130101); **B29C49/48** (20130101); B29K2067/003 (20130101); B29L2031/7158 (20130101)

Field of Classification Search

USPC: None

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This is a National Stage Application filed under 35 U.S.C. 371 based on International Patent Application No. PCT/CN2020/118769, filed on Sep. 29, 2020, which claims priority to Chinese Patent Application No. 202010979443.5 filed on Sep. 17, 2020 and claims priority to Chinese Patent Application No. 202022045751.9 filed on Sep. 17, 2020, disclosures of all of which are incorporated herein by reference in their entireties.
TECHNICAL FIELD
(2) The present application relate to the field of bottle blowing technologies, for example, a blow-molding mold, a bottle blowing machine and a plastic bottle.
BACKGROUND
(3) At present, plastic feeding bottles with self-contained handles and beverage bottles with self-contained handles on the market are mainly divided into two types, one type is a bottle with a self-contained handle and made of High Density Polyethylene (HDPE) material, and the other type is a bottle with a rear handle and made of a polyethylene glycol terephthalate (PET) material. The bottle with the handle and made of the HDPE material has the defects that sealing is incomplete, a bottle body is easy to damage, and a volume of the bottle cannot be controlled; the other bottle made of the PET material has the advantages of being good in sealing performance, firm in bottle body, high in oxygen barrier performance, transparent, easy to observe and the like compared with the bottle made of the HDPE material, but at present, the bottle made of the PET material needs to be provided with the rear handle, and the handle in the bottle with the rear handle and made of the PET material needs to be produced and installed through a specific equipment and is prone to falling off after being installed. Due to the above reasons, related enterprises in the industry want to replace the bottle with the handle and made of the PET material with the bottle with the rear handle and made of the PET material on the market at present.
(4) In order to ensure that a position of the handle is not changed and the shape is complete, a preform at a handle part of a bottle preform may only be transversely stretched and cannot be longitudinally stretched. Therefore, a technical treatment needs to be performed on the handle part in a bottle preform heating process, and a condition that in a bottle blowing process, the handle part softens after being too high in temperature and thus deforms, and further causes a bottle blowing failure is prevented. The bottle with a self-contained handle and made of the PET material is manufactured by using a two-step bottle blowing process method, that is, the bottle preform is firstly formed by injection molding, and then the bottle preform is blown into the bottle in a blow-

molding mold. Compared with a conventional bottle preform, the bottle preform with a self-contained handle has an additional assembly, i.e., the handle, so that problems which do not occur in a conventional bottle preform blowing process may be generated in a bottle preform blowing process. After the bottle preform is blown, a wrinkle is easily formed at a connection of the bottle shoulder opposite to the handle and the bottle neck, so that the quality of the bottle is unqualified.

SUMMARY

(5) The present application provides a blow-molding mold and a bottle blowing machine.

(6) The present application provides a plastic bottle.

(7) An embodiment provides a blow-molding mold. The blow-molding mold includes a mold cavity, where a bottle body, a bottle shoulder, a handle and a bottle neck are formed within the mold cavity after a bottle preform is blown, the bottle shoulder is located between the bottle body and the bottle neck, one end of the handle is connected to the bottle shoulder, and the other end of the handle is connected to the bottle shoulder or the bottle neck, the mold cavity includes a first concave region to have a raised step structure formed on the bottle neck.

(8) As an alternative to the present application, a cavity wall of the mold cavity is provided with at least one first rib to have a first groove formed on the bottle shoulder.

(9) As an alternative to the present application, the at least one first rib is disposed in a direction perpendicular to a height direction of the bottle shoulder.

(10) As an alternative to the present application, a cavity wall of the mold cavity is further provided with at least one second rib to have a non-closed second groove or an annularly closed second groove formed on the bottle body.

(11) As an alternative to the present application, the at least one second rib is disposed on a plane or the at least one second rib is disposed on the cavity wall in a wave shape.

(12) As an alternative to the present application, a number of second ribs is not less than two, and the second ribs are disposed at intervals or partially connected.

(13) As an alternative to the present application, the mold cavity includes a second concave region configured for accommodating the handle.

(14) An embodiment provides a bottle blowing machine. The bottle blowing machine includes the blow-molding mold of any one of the schemes described above.

(15) An embodiment provides a plastic bottle. The plastic bottle is made by the bottle blowing machine described above.

(16) As an alternative to the present application, the plastic bottle is made of PET.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a schematic view before blowing a bottle preform with a self-contained handle by using a conventional mold;

(2) FIG. **2** is a schematic view after blowing a bottle preform with a self-contained handle by using a conventional mold;

(3) FIG. **3** is an enlarged view at A of FIG. **2**;

(4) FIG. **4** is a schematic structural view of a mold cavity of a blow-molding mold according to an embodiment of the present application;

(5) FIG. **5** is an enlarged view at B of FIG. **4**;

(6) FIG. **6** is an enlarged view at C of FIG. **4**;

(7) FIG. **7** is a schematic perspective view of a plastic bottle according to an embodiment of the present application;

(8) FIG. **8** is a schematic structural view of a mold cavity of a blow-molding mold according to another embodiment of the present application;

(9) FIG. **9** is a schematic perspective view of a plastic bottle blown by using the blow-molding mold shown in FIG. **8**;
(10) FIG. **10** is a schematic structural view of a mold cavity of a blow-molding mold according to still another embodiment of the present application;
(11) FIG. **11** is a schematic perspective view of a plastic bottle blown by using the blow-molding mold shown in FIG. **10**;
(12) FIG. **12** is a schematic structural view of a mold cavity of a blow-molding mold according to yet another embodiment of the present application;
(13) FIG. **13** is a schematic perspective view of a plastic bottle blown by using the blow-molding mold shown in FIG. **12**; and
(14) FIG. **14** is a schematic perspective view of a blow-molding mold according to an embodiment of the present application.
(15) In the drawings: **1′**, Conventional mold; **1**, Blow-molding mold; **10**, Mold cavity; **11**, Left half mold; **12**, Right half mold; **13**, Bottom mold; **101**, First concave region; **102**, First rib; **103**, Second rib; **104**, Second concave region; **105**, Third rib; **100**, Bottle preform; **100′**, Plastic bottle; **110**, Bottle neck; **120**, Bottle shoulder; **1201**, Wrinkle; **130**, Handle; **140**, Bottle body; **200**, Stretching rod; **1100**, Step structure; **1200**, First groove; **1300**, Second groove; and **1400**, Third groove.
DETAILED DESCRIPTION
(16) A technical scheme of the present application will be further described in conjunction with the drawings and specific embodiments below.
(17) In the present application, unless expressly specified and limited otherwise, a first feature being "above" or "below" a second feature may include the first feature and the second feature being in direct contact, and may also include the first feature and the second feature not being in direct contact but being in contact through an additional feature between them. Moreover, the first feature being "above" the second feature includes the first feature being directly above and obliquely above the second feature, or simply represents that the first feature is at a higher level than the second feature. The first feature being "below" the second feature includes the first feature being directly below and obliquely below the second feature, or simply represents that the first feature is at a lower level than the second feature.
(18) Moreover, the terms "first" and "second" are used for descriptive purposes only and are not to be construed as indicating or implying relative importance or as implicitly indicating a number of technical features indicated. Thus, features defined as "first" or "second" may explicitly or implicitly include one or more such features.
(19) According to a two-step bottle blowing process, a bottle preform **100** needs to be manufactured firstly, and then a specific mold is used for blowing the bottle preform **100**, as shown in FIGS. **1** to **3**. When a conventional mold **1′** is used for blowing a bottle preform **100** with a self-contained handle **130**, due to a fact that the handle **130** is additionally provided, in a stretching and expanding process of the bottle preform **100**, a preform at the bottle body **140** part is expanded firstly and fills a lower portion of a mold cavity **10**, and at this time, the handle **130** part of the bottle preform **100** is not attached to a cavity wall of the mold cavity **10**, and along with blowing, a preform at the bottle body **140** part may push the handle **130** to the cavity wall of the mold cavity **10** under the action of compressed air, and finally the preform and the handle **130** are attached to the cavity wall of the mold cavity **10** together under the action of the compressed air, after a blowing is completed, a wrinkle **1201** (as shown in FIG. **3**) is prone to occurring at a connection of the bottle shoulder **120** and the bottle neck **110**, thereby a quality defect is caused.
(20) As shown in FIGS. **4** to **7**, the blow-molding mold **1** in this embodiment includes the mold cavity **10**, where the bottle neck **110**, the bottle shoulder **120**, the handle **130** and the bottle body **140** are formed within the mold cavity **10** after the bottle preform **100** is blown, the bottle shoulder **120** is located between the bottle neck **110** and the bottle body **140**, one end of the handle **130** is connected to the bottle shoulder **120**, the other end of the handle **130** is connected to the bottle neck

110, and two ends of the handle **130** may also be connected to the bottle shoulder **120**. The cavity wall of the mold cavity **10** is provided with a first concave region **101** to have a raised step structure **1100** formed on the bottle neck **110**.
(21) According to the blow-molding mold **1** of this embodiment, the cavity wall of the mold cavity **10** is provided with a first concave region **101** to have the raised step structure **1100** formed on the bottle neck **110**, so that more preforms are occupied at this position, thereby a wrinkle **1201** occurring at the connection of the bottle shoulder **120** and the bottle neck **110** is prevented, and the blowing quality of the bottle preform **100** is improved.
(22) Optionally, the cavity wall of the mold cavity **10** is provided with at least one first rib **102** to have a first groove **1200** formed on the bottle shoulder **120**. The first rib **102** may limit a preform at the bottle shoulder **120** part from moving towards the bottle neck **110**, so that the probability of occurrence of wrinkles **1201** is reduced, and the first groove **1200** may also play a role in enhancing the structural strength of the bottle shoulder **120**. Due to the presence of the handle **130**, the first rib **102** cannot form a closed annular shape, and thus the first groove **1200** is not in a closed configuration.
(23) Exemplarily, the first rib **102** is disposed in a direction perpendicular to a height direction of the bottle shoulder **120**, so that the first rib **102** can better block the preform at the bottle shoulder **120** part from moving towards the bottle neck **110**.
(24) Optionally, the cavity wall of the mold cavity **10** is further provided with at least one second rib **103** to have a non-closed or annularly closed second groove **1300** formed on the bottle body **140**. Optionally, an arrangement position of the second rib **103** is facing the bottle shoulder **120**, so that the second rib **103** can limit the preform at the bottle body **140** part from moving towards the bottle shoulder **120** and the handle **130**, the bottle shoulder **120** is prevented from wrinkling, and the handle **130** is prevented from shifting due to the movement of the preform. Moreover, the second groove **1300** also plays a role in enhancing the structural strength of the bottle body **140**, and multiple second groove **1300** may be obtained by providing multiple second ribs **103**, so that the structural strength of the bottle body **140** is greatly improved, and thus the bottle body **140** is not easy to deform.
(25) Exemplarily, the second rib **103** is disposed on a same plane or the second rib **103** is disposed on the cavity wall of the mold cavity **10** in a wave shape, and a specific shape of the second rib **103** may be flexibly selected according to actual needs. As shown in FIG. **8** and FIG. **9**, the second rib **103** has a closed ring shape or a non-closed arc shape, and the second ribs **103** are disposed on a same plane so that heights of the second groove **1300** are the same everywhere. As shown in FIGS. **10** and **11**, the second rib **103** exhibits a wave shape, so that the second groove **1300** also exhibit a wave shape, and the second grooves **1300** are not at a same height throughout.
(26) When a number of the second ribs **103** is larger than two, the second ribs **103** may be disposed at intervals and may be partially connected together. As shown in FIGS. **12** and **13**, the second ribs **103** are partially connected, so that a part of the second grooves **1300** is overlapped.
(27) As shown in FIG. **4**, optionally, the mold cavity **10** includes a second concave region **104** configured for accommodating the handle **130**. A shape of the second concave region **104** matches a shape of the handle **130** so that the handle **130** may be stopped in a correct position during the blowing process of the bottle preform **100**.
(28) As shown in FIGS. **4** and **7**, in some embodiments, the cavity wall of the mold cavity **10** is provided with a third rib **105** to have a third groove **1400** formed at a connection of the bottle shoulder **120** and the bottle body **140**. The third rib **105** functions similar to the second rib **103**, and may also limit the preform at the bottle body **140** part from moving towards the bottle shoulder **120**, thereby preventing the bottle shoulder **120** from wrinkling.
(29) Optionally, as shown in FIG. **14**, the blow-molding mold **1** includes a left half mold **11**, a right half mold **12** and a bottom mold **13**, where the left half mold **11**, the right half mold **12** and the bottom mold **13** are assembled to form the mold cavity **10**. In combination with FIGS. **1** and **2**, the

bottle preform **100** is blown in the following steps: firstly, the left half mold **11** is separated from the right half mold **12**, then the heated bottle preform **100** is placed between the left half mold **11** and the right half mold **12**, then the left half mold **11**, the right half mold **12** and the bottom mold **13** are closed, and a stretching rod **200** is inserted into the bottle preform **100** from a preform opening to longitudinally stretch the bottle preform **100**, and meanwhile, compressed air is blown into the bottle preform **100** from the preform opening to transversely stretch the bottle preform **100**, and after the bottle preform **100** is blown, the blow-molding mold **1** is opened.
(30) An embodiment of the present application further provides a bottle blowing machine, the bottle blowing machine includes the blow-molding mold **1** of any one of the embodiments described above, so that the bottle blowing machine may perform the high-quality blowing on the bottle preform **100** with a self-contained handle **130**.
(31) As shown in FIG. **7**, FIG. **9**, FIG. **11** and FIG. **13**, an embodiment of the present disclosure further provides a plastic bottle **100′**, the plastic bottle **100′** is made by the bottle blowing machine described above, the plastic bottle **100′** has a self-contained handle **130**, instead of being in the form of an independent handle which is additionally installed after blowing, whereby the production process is simplified, the production cost is reduced, and a problem that the independent handle falls off is not worried.
(32) Optionally, the plastic bottle **100′** is made of PET, so that the plastic bottle **100′** has the advantages of being good in sealing performance, firm in bottle body, high in oxygen barrier performance, transparent, easy to observe and the like.
(33) As a preferred implementation of the present application, in the description of this specification, a description referring to the term “preferred” and the like means that a specific feature, structure, material, or characteristic described in conjunction with the embodiment or example is included in at least one embodiment or example of the present application. In the present specification, schematic recitations of the above terms do not necessarily refer to the same embodiment or example. Moreover, the specific features, structures, materials, or characteristics described may be combined in a suitable manner in any one or more embodiments or examples.

Claims

1. A blow-molding mold, comprising a mold cavity, wherein a bottle body, a bottle shoulder, a handle and a bottle neck are formed within the mold cavity after a bottle preform is blown, the bottle shoulder is located between the bottle body and the bottle neck, one end of the handle is connected to the bottle shoulder, and another end of the handle is connected to the bottle neck, the mold cavity is provided with a first concave region to have a raised step structure formed on the bottle neck below the other end of the handle.
2. The blow-molding mold of claim 1, wherein a cavity wall of the mold cavity is provided with at least one first rib, to have at least one first groove formed on the bottle shoulder.
3. The blow-molding mold of claim 2, wherein the at least one first rib is disposed in a direction perpendicular to a height direction of the bottle shoulder.
4. The blow-molding mold of claim 1, wherein a cavity wall of the mold cavity is further provided with at least one second rib to have a non-closed second groove or an annularly closed second groove formed on the bottle body.
5. The blow-molding mold of claim 4, wherein the at least one second rib is disposed on a same plane or the at least one second rib is disposed on the cavity wall in a wave shape.
6. The blow-molding mold of claim 4, wherein a number of second ribs is not less than two, and the second ribs are disposed at intervals or partially connected.
7. The blow-molding mold of claim 1, wherein the mold cavity comprises a second concave region configured for accommodating the handle.
8. A bottle blowing machine, comprising a blow-molding mold, wherein the blow-molding mold

comprises a mold cavity, wherein a bottle body, a bottle shoulder, a handle and a bottle neck are formed within the mold cavity after a bottle preform is blown, the bottle shoulder is located between the bottle body and the bottle neck, one end of the handle is connected to the bottle shoulder, and another end of the handle is connected to the bottle neck, the mold cavity is provided with a first concave region to have a raised step structure formed on the bottle neck below the other end of the handle.

9. A plastic bottle, made by a bottle blowing machine comprising a blow-molding mold, wherein the blow-molding mold comprises a mold cavity, wherein a bottle body, a bottle shoulder, a handle and a bottle neck are formed within the mold cavity after a bottle preform is blown, the bottle shoulder is located between the bottle body and the bottle neck, one end of the handle is connected to the bottle shoulder, and another end of the handle is connected to the bottle neck, the mold cavity is provided with a first concave region to have a raised step structure formed on the bottle neck below the other end of the handle.

10. The plastic bottle of claim 9, wherein the plastic bottle is made of polyethylene glycol terephthalate (PET).

11. The bottle blowing machine of claim 8, wherein a cavity wall of the mold cavity is provided with at least one first rib, to have at least one first groove formed on the bottle shoulder.

12. The bottle blowing machine of claim 11, wherein the at least one first rib is disposed in a direction perpendicular to a height direction of the bottle shoulder.

13. The bottle blowing machine of claim 8, wherein a cavity wall of the mold cavity is further provided with at least one second rib to have a non-closed second groove or an annularly closed second groove formed on the bottle body.

14. The bottle blowing machine of claim 13, wherein the at least one second rib is disposed on a same plane or the at least one second rib is disposed on the cavity wall in a wave shape.

15. The bottle blowing machine of claim 13, wherein a number of second ribs is not less than two, and the second ribs are disposed at intervals or partially connected.

16. The bottle blowing machine of claim 8, wherein the mold cavity comprises a second concave region configured for accommodating the handle.

17. The plastic bottle of claim 9, wherein a cavity wall of the mold cavity is provided with at least one first rib, to have at least one first groove formed on the bottle shoulder.

18. The plastic bottle of claim 17, wherein the at least one first rib is disposed in a direction perpendicular to a height direction of the bottle shoulder.

19. The plastic bottle of claim 9, wherein a cavity wall of the mold cavity is further provided with at least one second rib to have a non-closed second groove or an annularly closed second groove formed on the bottle body.

20. The plastic bottle of claim 19, wherein the at least one second rib is disposed on a same plane or the at least one second rib is disposed on the cavity wall in a wave shape.